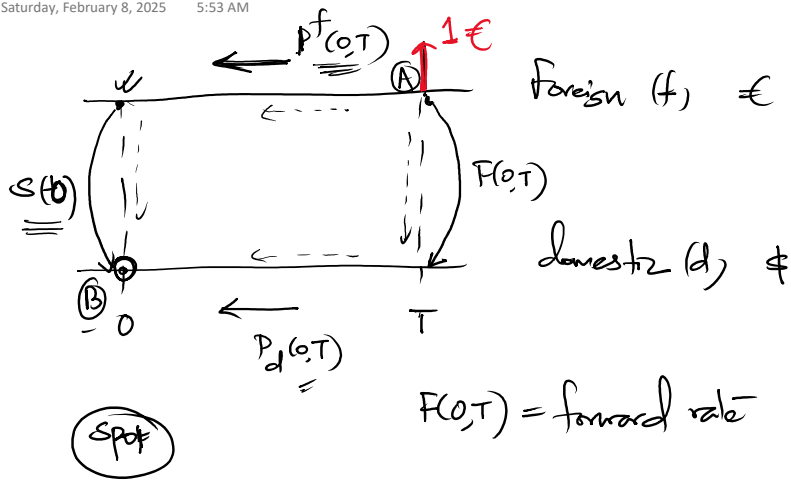




$$1 \times P_f(0,T) > S(0) \text{ \$}$$

$$= 1 \times F(0,T) \times P_d(0,T) \text{ \$}$$

(no arbitrage)

$$F(0,T) = S(0) \times \frac{P_f(0,T)}{P_d(0,T)}$$

(IRP)

$$F(0,T) = S(0) \cdot e^{(r_d - r_f)T}$$

$r_d, r_f = \text{Continuous compounding}$

$$F(0,T) = S(0) \times \left(\frac{1 + i_d T}{1 + i_f T} \right)$$

$i_d, i_f = \text{simple compounding}$

$$T = \frac{D}{360}$$

$$e^x \approx 1+x$$

$$F(0,T) = S(0) e^{(r_d - r_f)T}$$

$$\approx S(0) (1 + (r_d - r_f)T)$$

$$F(0,T) = S(0) \left(\frac{1 + i_d T}{1 + i_f T} \right)$$

$$F - S = S(0) \left[\frac{(1 + i_d T) - (1 + i_f T)}{1 + i_f T} \right] = \frac{S(0) \times (i_d - i_f) \frac{D}{360}}{1 + i_f \times \frac{D}{360}} \times 10000$$

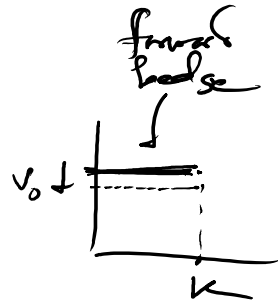
option hedge (Put)

at T,
$$\underline{N_f \times S(T)} + \underline{N_f \times (k - S_T)^+} - \underline{V_0}$$

Case I, $S(T) < k$

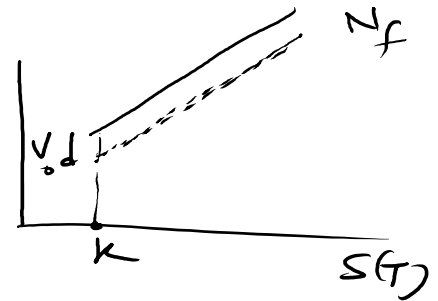
$$N_f \times S(T) + N_f \times (k - S_T) - V_0$$

$$= \underline{N_f \times k - V_0} \rightarrow \text{Constant}$$



Case II, $S_T > k$

$$\underline{N_f \times S(T)} - \underline{V_0}$$



$$P + S = C + Ke^{-rT}$$

$$\frac{\partial P}{\partial S} + 1 = \frac{\partial C}{\partial S} \longrightarrow \frac{\partial^2 P}{\partial S^2} = \frac{\partial^2 C}{\partial S^2}$$

$$\Delta_P = \Delta_C - 1$$

$$\gamma_P = \gamma_C$$

$$\downarrow$$

$$N(d_1)$$

$$\delta_C = \underline{N(d_1)} - 1$$

$$\delta_P = \underline{N(d_1)} - 1$$

$$\text{portfolio delta} = \delta^{\text{port}}$$

$$\text{forward delta} = -N_f e^{-r_f T}$$

$$\delta^{\text{port}} - N_f e^{-r_f T} = 0$$

$$\Rightarrow \underline{N_f} = \frac{\delta^{\text{port}}}{e^{-r_f T}}$$

$$\text{option delta} = N_f (N(d_1) - 1) e^{-r_f T}$$

$$\delta^{\text{port}} + N_f (N(d_1) - 1) e^{-r_f T} = 0$$

$$\Rightarrow N_f = \frac{\delta^{\text{port}}}{(1 - N(d_1)) e^{-r_f T}}$$